Coding & Solution

Coding & Solution This section describes the implementation quality and completeness of the ComicCraft solution. The project is a web-based AI comic generation application built with FastAPI, Google Gemini models, Stable Diffusion, HTML/CSS/Jinja2, and FPDF.

Instructions / Implementation Alignment:

• The application is organized into separate modules for story outline generation, story narration, image generation, layout building, PDF export, routing, and frontend templates.

• The solution addresses the core problem of transforming user prompts into personalized comic stories, illustrations, and a downloadable PDF.

• The project requires Python environment setup, package installation, environment variables for API credentials, and the FastAPI/Uvicorn server.

• The documented project structure includes backend Python modules, Jinja2 HTML templates, static panel/export folders, and configuration/dependency files.

Solution Summary

Key Source Files / Modules

Code Quality Checklist

Additional Notes / Comments:

• The source project document identifies FastAPI, Gemini Flash, Gemini Pro, Stable Diffusion, Hugging Face Diffusers, Jinja2, and FPDF as core technologies.

• The documented workflow covers model selection, architecture, core functionality development, routes.py development, frontend development, local deployment, and testing.

• The current version focuses on single comic generation without user account management; future enhancements include user profiles, comic libraries, and multi-page story arcs.

• Repository URL and actual version-control status were not provided in the source material, so they are marked as not specified rather than invented.

Team ID: SWTID-2026-3296

Date	29 September 2026
Team ID	SWTID-2026-3296
Project Name	ComicCraft - AI Comic Story Creator using Gemini Models
Maximum Marks	5 Marks

Field	Details
Repository Link / URL	Not provided in the source document
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	FastAPI, Jinja2
AI / Model Integration	Google Gemini Flash, Google Gemini Pro, Stable Diffusion via Hugging Face Diffusers
PDF / Export Technology	FPDF
Key Features Implemented	5-panel outline generation; narration and dialogue generation; AI image generation; comic layout; web preview; PDF export; FastAPI browser and JSON API routes
Pending / Incomplete Features	User account management, user profiles, comic libraries, and multi-page story arcs are described as future enhancements
Setup / Run Instructions	Create and activate a Python virtual environment; install required packages; configure GEMINI_API_KEY and HF_API_KEY; run `uvicorn app.main:app --reload`; access the application at http://127.0.0.1:8000 and API docs at /docs

Source File / Module	Purpose
gemini_flash.py	generate_outline() creates the structured 5-panel comic outline.
gemini_pro.py	generate_story() creates narration and character dialogue.
image_generator.py	generate_image() creates comic-style illustrations using Stable Diffusion.
layout_builder.py	build_comic_layout() organizes images and story content into panel layout.
exporters.py	save_pdf() compiles the comic into a multi-page PDF using FPDF.
routes.py	Handles FastAPI routes, user input processing, AI workflow integration, and error handling.
index.html	Collects story prompt, character name, setting, tone, and art style.

Source File / Module	Purpose
comic_preview.html	Displays generated comic panels, images, descriptions, captions, and narration.
export_success.html	Confirms successful comic PDF export.

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Not specified in source